Given the following system of equations

ax+by=e
cx+dy=f

where x and y are unknowns and a,b,c,d,e and f are constants
a solution can be shown to be

x= 1/det (de-bf)
y= 1/det (-ce+af)

where det = ad-bc (which must not = 0)

write a program that reads the 6 values a thru f and then outputs the
solution.

Test your program using this system of equations

2x-3y=-4
5x+y=7

Below is a simple Python program that reads the six constants a, b, c, d, e, f, computes the determinant, and prints the solution x and y:

python   Copy code

```python
# Solve system:
#   ax + by = e
#   cx + dy = f

a = float(input("Enter a: "))
b = float(input("Enter b: "))
c = float(input("Enter c: "))
d = float(input("Enter d: "))
e = float(input("Enter e: "))
```

```python
f = float(input("Enter f: "))

det = a*d - b*c

if det == 0:
    print("Determinant is zero — no unique solution.")
else:
    x = (d*e - b*f) / det
    y = (-c*e + a*f) / det

    print("Solution:")
    print("x =", x)
    print("y =", y)
```

✔ Test the program with the given system:

You provided:

$$2x - 3y = -4$$

$$5x + y = 7$$

So:

- $a = 2$
- $b = -3$
- $c = 5$
- $d = 1$
- $e = -4$
- $f = 7$

Compute:

Determinant:

$$\det = ad - bc = (2)(1) - (-3)(5) = 2 + 15 = 17$$

Now

$$x = \frac{de - bf}{\det} = \frac{(1)(-4) - (-3)(7)}{17} = \frac{-4 + 21}{17} = \frac{17}{17} = 1$$

$$y = \frac{-ce + af}{\det} = \frac{-(5)(-4) + 2(7)}{17} = \frac{20 + 14}{17} = \frac{34}{17} = 2$$

✅ **Final Answer:**

$$x = 1, \quad y = 2$$

Let me know if you'd like the program in another language (C, Java, etc.)!